

# Change record — P1 Core Infrastructure Build

Retroactive change record, written to the standard Halden uses for every change (feeds the change log and the P10 governance dashboard). In a small company the same person often proposes and approves; that separation is documented honestly rather than pretended.

| Field                  | Value                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------|
| Change ID              | CHG-2026-001                                                                                     |
| Title                  | P1 — Core infrastructure build: AD DS, DNS, DHCP, file services, GPO baseline, Linux integration |
| Raised by              | IT (Muhammad Ibrahim Akmal)                                                                      |
| Approved by            | Managing Director ( <i>business owner of the fictional Halden; see note on the lab</i> )         |
| Date raised            | 2026-09-29                                                                                       |
| Planned implementation | 2026-10-02 onwards, phase by phase (Phase 0 → Phase 8)                                           |
| Category / risk        | Infrastructure — High risk (Tier 0: authentication, DNS, DHCP)                                   |
| Affected services      | Logon, DNS, DHCP, all file shares, Group Policy, Linux SSH access                                |
| Affected systems       | HOST01 (Hyper-V), FW01, DC01, DC02, FS01, LNX01, WS01, WS02                                      |

## 1. Description of the change

Replace the single aging server with a redundant, documented Windows Server 2025 foundation: two replicating domain controllers with AD-integrated DNS, a DHCP failover pair, a dedicated file server with DFS namespace, access-based enumeration, FSRM quotas and file screens and shadow copies, a tiered Group Policy baseline, and one Ubuntu server joined to the domain with group-controlled SSH and sudo. All of it is created by idempotent, logged scripts kept in the repository, plus an as-built document, a permission matrix and user runbooks.

## 2. Reason for the change

The current single server is a single point of failure for logon, addressing and files, and all 85 staff hold Full Control on the shared drive so a single mistake or a ransomware infection can destroy any department's data. There is also no written answer to "who can read Finance data" and no self-service file restore. The change removes the single point of failure, applies least privilege by role, and produces the documentation and evidence that the business currently lacks.

## 3. Impact and risk

| Area  | Impact                                                                                                   | Mitigation                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Users | New drive letters appear; a screen lock is enforced; folders that a user has no right to open are hidden | 1-page brief to all staff in advance; Service Desk briefed; ticket route for missing access |

| Area               | Impact                                                           | Mitigation                                                                                               |
|--------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Logon / DNS / DHCP | Outage would stop the whole company                              | Work done out of hours, phase by phase; DC02 and DHCP failover tested by powering DC01 off               |
| Data               | New permission model could lock a user out of a folder they need | Access approved by department heads first; permission matrix kept; Administrators retain recovery access |
| Access control     | A mistake could grant too much access                            | Only <code>DL_</code> groups ever appear on ACLs; automated ACL audit must report <b>0 violations</b>    |

## 4. Implementation plan (phases)

| Phase | Work                                                          | Window       | Verification                                       |
|-------|---------------------------------------------------------------|--------------|----------------------------------------------------|
| 0     | Design document (IP plan, naming, OU tree, DNS/DHCP design)   | 2026-10-02   | Design reviewed against the plan                   |
| 1     | DC01 promoted; DNS zones, forwarders, scavenging, time source | Out of hours | <code>dcdiag /q</code> clean; DNS records resolve  |
| 2     | DC02 promoted; sites/subnets; replication verified            | Out of hours | <code>repadmin /replsummary</code> 0 fails         |
| 3     | DHCP scope, options, 50/50 failover, name protection          | Out of hours | Failover state Normal; client renews with DC01 off |
| 4     | OU tree, AGDLP groups, 85 users imported from CSV             | Out of hours | Import log; idempotent second run                  |
| 5     | FS01 shares, ACLs, DFS-N, ABE, FSRM, shadow copies            | Out of hours | ACL audit 0 violations                             |
| 6     | GPO baseline + Central Store + GPO backup                     | Out of hours | <code>gpresult /h</code> on WS01                   |
| 7     | LNK01 joined; AD-based SSH and sudo                           | Out of hours | Denied for non-members; sudo for members           |
| 8     | As-built doc, diagram, runbooks                               | Same evening | Documents produced from live output                |

## 5. Test plan (acceptance)

1. Power off DC01 — logons, DNS resolution and DHCP leases continue via DC02.
2. Finance user opens the company share — the HR folder is not visible.
3. Sales user copies an `.exe` into the Sales share — blocked by the file screen.
4. Delete a file — restore it through "Previous Versions" without IT.
5. Non-IT domain user connects to LNK01 by SSH — denied.
6. Re-run the user import script — no duplicates, no errors (idempotent).

## 6. Backout plan

Each phase is snapshotted before it runs ( `snap-p1-ph<N>-before` ). Backout for any phase is to power off the affected VM, revert the snapshot and re-run the previous phase's scripts, which are idempotent. Domain controllers are fixed forward where possible: reverting a DC snapshot can disturb replication, so it is only used as a last resort and is recorded in `DECISIONS.md` .

## 7. Post-implementation review

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To be completed when the last phase is verified: actual completion date, acceptance test results, anything that went wrong, and whether any phase overran. Results are recorded in `projects/p01-core-infrastructure/README.md` with a source file for every number.

Note: this is a home-lab change record for a fictional company (Halden Distribution Ltd.). The technical content, scripts and test results are real; the business approval line represents the lab owner's decision, not a real customer sign-off.